Power System Economics: Answers

Question 1 Solution

a. The maximum consumption of the consumer corresponds to the point where the price is equal to zero. This means:

$$p = 0 \Rightarrow -10 * D + 2000 = 0 \Rightarrow D\max = 200[\text{units}] \quad (1)$$

The maximum price the consumer is prepared to pay corresponds to the point where the demand is equal to zero. This means:

$$D = 0 \Rightarrow p\max = 2000[\text{£/unit}] \quad (2)$$

Given that the consumer's surplus is given by the area under the demand curve and above the market price, the maximum consumer's surplus corresponds to the point where the price is equal to zero. For this price, this area is equal to

$$CS\max = 0.5 * D\max * p\max = 200{,}000[\text{£}] \quad (3)$$

The gross benefit of the consumer is given by the integral of the marginal benefit function which is identical to the inverse demand function. This means (assuming there is no constant benefit term):

$$GB = \int -10 * D + 2000 = -5 * D^2 + 2000 * D[\text{£}] \quad (4)$$

The above gross benefit can be expressed as a function of price, if we express D as a function of p:

$$p = -10 \cdot D + 2000 \Rightarrow D = 200 - 0.1 * p[\text{units}] \quad (5)$$

By inserting (5) into (4) we get:

$$GB = -5 * (200 - 0.1 * p)^2 + 2000 * (200 - 0.1 * p) = -0.05 * p^2 + 200000[\text{£}] \quad (6)$$

b. The average cost is given by the cost divided by the production level:

$$AC = C / Q = \frac{5}{2} * Q + 200[\text{£/unit}] \quad (7)$$

Answers

The marginal cost is given by the derivative of the cost:

$$MC = 5 * Q + 200 [£/\text{unit}] \quad (8)$$

The gross revenue is given by the product of the marginal cost times the production level:

$$GR = MC * Q = 5 * Q^2 + 200 * Q [£] \quad (9)$$

The profit (or net revenue) is given by the difference between the gross revenue and cost:

$$NR = GR - C = 2.5 * Q^2 [£] \quad (10)$$

c. The equilibrium demand level is calculated by equalizing the marginal benefit of the consumer with and the marginal cost of the producer:

$$MB = MC \Rightarrow -10 * D + 2000 = 5 * D + 200 \Rightarrow Deq = 120 [\text{units}] \quad (11)$$

The equilibrium price is calculated by replacing the equilibrium demand level in the inverse demand function:

$$peq = -10 * Deq + 2000 = 800 [£/\text{unit}] \quad (12)$$

d. The consumer's gross benefit is given by inserting (11) into (4):

$$GBeq = 168000 [£] \quad (13)$$

The consumer's surplus is given by the difference between the gross benefit and the consumer's charges:

$$CSeq = GBeq - peq * Deq = 72000 [£] \quad (14)$$

The producer's average cost is calculated by inserting (11) into (7):

$$ACeq = 500 [£/\text{unit}] \quad (15)$$

The gross revenue is calculated by inserting (11) into (9):

$$G\,\text{Re}\,q = 96000 [£] \quad (16)$$

The profit (or net revenue) is calculated by inserting (11) into (10):

$$N\,\text{Re}\,q = 36000 [£] \quad (17)$$

Answers

The social welfare is given by the sum of consumer's surplus and producer's profit:

$$SWeq = CSeq + N\,\mathrm{Re}\,q = 108000\,[£] \tag{18}$$

e. An increase of the price by 20% means that the new price will be:

$$p = 1.2 * peq = 960[£/\text{unit}] \tag{19}$$

The new demand level is calculated by replacing (19) in the inverse demand function:

$$960 = -10 * D + 2000 \Rightarrow D = 104\,[\text{units}] \tag{20}$$

The new gross benefit of the consumer is calculated by inserting (20) into (4):

$$GB = 153920\,[£] \tag{21}$$

The new consumer's surplus is given by the difference between the new gross benefit (21) and the new consumer's charges:

$$CS = GB - p * D = 54080\,[£] \tag{22}$$

By comparing (14) with (22), we conclude that this price increase is not beneficial for the consumer.

The new profit (or net revenue) of the producer is given by the difference between the new producer payment and the new producer cost:

$$NR = 52000\,[£] \tag{23}$$

By comparing (17) with (23), we conclude that this price increase is beneficial for the producer.

The new social welfare is given by the sum of the new consumer's surplus and the new producer's profit:

$$SW = CS + NR = 106080\,[£] \tag{24}$$

By comparing (18) with (24), we conclude that this price increase is not beneficial for the society overall.

Answers

Question 2 Solution

a) In perfect markets, individual actions have no effect on the market price and all participants are viewed as passive "price-takers". However, in real (imperfect) competition, producers are able to exercise market power by reducing their output or raising their offer price, in order to change the market price and increase their profit. In this case, each company must consider possible actions of others, since they have a direct effect on the company's profit.

b)

(i) Marginal cost of A: 20£/MWh
Marginal cost of B: 30 £/MWh
Generating company A will set its price just below 30 £/MWh. Generating company B cannot set its price below 30 £/MWh because it will be making a loss. As a result, all the demand will be met by Generating company A.
The demand will be: $D = 100 - p = 100 - 30 = 70\,MW$

Hence, $\begin{matrix} P_A = 70\,MW \\ P_B = 0\,MW \end{matrix}$ with market price = 30 £/MWh.

The profit made by company B is zero.
The cost of company A is: $20£/MWh \times 70MW = 1400£/h$
The revenue of company A is: $30£/MWh \times 70MW = 2100£/h$
Hence, the profit made is: $2100£/h - 1400£/h = 700£/h$

(ii) The conditions at the equilibrium state are:

$$\frac{d\Omega_A}{dP_A} = 0 \qquad (1)$$

$$\frac{d\Omega_B}{dP_B} = 0 \qquad (2)$$

Given that the market price is given by:

$$p = 100 - D = 100 - P_A - P_B \qquad (3)$$

The profit of the two companies is calculated as:

$$\Omega_A = p * P_A - C_A \xrightarrow{(3)} \Omega_A = -P_A^2 + 80 * P_A - P_A * P_B \qquad (4)$$

$$\Omega_B = p * P_B - C_B \xrightarrow{(3)} \Omega_B = -P_B^2 + 70 * P_B - P_A * P_B \qquad (5)$$

By substituting (4) and (5) into (1) and (2) respectively and solving the resulting system of equations we get:

Answers

$$P_A = 30MW \quad (6)$$

$$P_B = 20MW \quad (7)$$

The market price is calculated by substituting (6)-(7) into (3):

$$p = 50£/MWh \quad (8)$$

The profits are calculated by substituting (6)-(7) into (4)-(5):

$$\Omega_A = 900£/h \quad (9)$$

$$\Omega_B = 400£/h \quad (10)$$

(iii) The major difference between Cournot and Bertrand competition is that the first models imperfect competition by introducing competition on prices. The latter introduces competition on quantity produced.

Answers

Question 3 Solution

a. Unconstrained optimal dispatch:

We need $MC_A = MC_B \Rightarrow 16 = 11 + 0.1 \cdot P_B \Rightarrow P_B = \frac{16-11}{0.1} = 50\,MW$

We also need $P_A + P_B = 310\,MW$

$P_A = 260MW$

$P_B = 50MW$

b. The cost functions can be found by integrating the marginal cost expressions (we assume that the integration constant is equal to 0).
The hourly cost of the unconstrained dispatch is:

$C_A = 16 \cdot P_A \Rightarrow C_A = 16 \cdot 260 = 4160£/h$

$C_B = 11 \cdot P_B + \frac{1}{2} \cdot 0.1 \cdot P_B{}^2 \Rightarrow 11 \cdot 50 + \frac{1}{2} \cdot 0.1 \cdot 50^2 = 675£/h$

$Total\,Cost = 4160£/h + 675£/h = 4835£/h$

c. We can use the following equations:

$P_A = F_{12} + F_{13}$

$P_B = -F_{12} + F_{23}$

Since all the transmission lines have the same impedance, we can also say:

$F_{12} + F_{23} = F_{13}$

Hence,

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} F_{1-2} \\ F_{1-3} \\ F_{2-3} \end{bmatrix} = \begin{bmatrix} P_A \\ P_B \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} F_{1-2} \\ F_{1-3} \\ F_{2-3} \end{bmatrix} = \begin{bmatrix} 70 \\ 190 \\ 120 \end{bmatrix}$$

d. Since this is the unconstrained dispatch (no constraints on flows), the marginal cost is the same at every bus:

$MC_A = 16£/MWh$, can be supplied directly from Generator A or B.

$MC_B = 16£/MWh$, can be supplied directly from Generator B or A.

$MC_c = 16£/MWh$, can be supplied from Generator A or B.

e. When 1MW is generated at Bus 1, 0.67 MW flow from Bus 1 to Bus 3, 0.33 MW flow from Bus 1 to Bus 2 and the same from Bus 2 to Bus 3.
When 1MW is generated at Bus 2, 0.67 MW flow from Bus 2 to Bus 3, 0.33 MW flow from Bus 2 to Bus 1 and the same from Bus 1 to Bus 3.
This means that in order to alleviate the overload of (190-160)=30 MW, we will have to reduce generation at Bus 1 by $\frac{30}{0.33} = 90MW$ and increase generation at Bus 2 by the same amount to maintain system power balance. The new dispatch should be:

$P_A = 260 - 90 = 170MW$

$P_B = 50 + 90 = 140MW$

Answers

f. Hourly cost of constrained dispatch:

$C_A = 16 \cdot P_A \Rightarrow C_A = 16 \cdot 170 = 2720£/h$

$C_B = 11 \cdot P_B + \frac{1}{2} \cdot 0.1 \cdot P_B{}^2 \Rightarrow 11 \cdot 140 + \frac{1}{2} \cdot 0.1 \cdot (140)^2 = 2520£/h$

$Total\,Cost = 2720£/h + 2520£/h = 5240£/h$

Hourly cost of security:

$5240£/h - 4835£/h = 405£/h$

g. The next MWh at node 1 can be directly supplied by generator A. This means that $MC_1 = 16£/MWh$

The next MWh at node 2 can be directly supplied by generator B. This means that $MC_2 = 11 + 0.1 \cdot 140 = 25£/MWh$.

In order to calculate the changes in dispatch required to supply the next MWh at node 3 we need to solve the following system of equations:

$\Delta P_A + \Delta P_B = 1$ (to ensure than an extra MWh is supplied to the system)

$0.67 * \Delta P_A + 0.33 * \Delta P_B = 0$ (to ensure that line from Bus 1 to Bus 3 is not overloaded)

Solution of this system gives:

$\Delta P_A = -1MW$

$\Delta P_B = 2MW$

This means that:

$MC_3 = 2 * MC_2 - MC_1 = 34£/MWh$

Answers

Question 4 Solution

a. The way to find the optimal dispatch and marginal prices is to create a set of equations: one stating that the sum of generation should be equal to the load and another equation stating that the marginal prices of the two generators should be the same.

Northern region:

$MC_1 = MC_2 \Rightarrow 4 + 0.02 \cdot P_1 = 5 + 0.03 \cdot P_2$

$P_1 + P_2 = 100\,MW$

Hence,

$P_1 = 80MW$

$P_2 = 20MW$

$MC_1 = MC_2 = 5.6£/MWh$

Payments to Generator 1: $P_1 \times MC_1 = 80MW \times 5.6£/MWh = 448£/h$

Payments to Generator 2: $P_2 \times MC_2 = 20MW \times 5.6£/MWh = 112£/h$

Demand charges: $Demand \times MC = 100MW \times 5.6£/MWh = 560£/h$

Southern region:

$MC_3 = MC_4 \Rightarrow 5 + 0.025 \cdot P_3 = 7 + 0.025 \cdot P_4$

$P_3 + P_4 = 400\,MW$

Hence,

$P_3 = 240MW$

$P_4 = 160MW$

$MC_3 = MC_4 = 11£/MWh$

Payments to Generator 3: $P_3 \times MC_3 = 240MW \times 11£/MWh = 2640£/h$

Payments to Generator 4: $P_4 \times MC_4 = 160MW \times 11£/MWh = 1760£/h$

Demand charges: $Demand \times MC = 400MW \times 11£/MWh = 4400£/h$

The marginal value of transmission is

$\pi_{SR} - \pi_{NR} = 11£/MWh - 5.6£/MWh = 5.4£/MWh$.

b.

(i) We need to find the optimal dispatch for the case C= 50MW (50MW moving from Northern region to Southern region).

Northern region:

$MC_1 = MC_2 \Rightarrow 4 + 0.02 \cdot P_1 = 5 + 0.03 \cdot P_2$

$P_1 + P_2 = 100MW + 50\,MW$

Hence,

$P_1 = 110MW$

$P_2 = 40MW$

$MC_1 = MC_2 = 6.2£/MWh$

Answers

Payments to Generator 1: $P_1 \times MC_1 = 110MW \times 6.2£/MWh = 682£/h$

Payments to Generator 2: $P_2 \times MC_2 = 40MW \times 6.2£/MWh = 248£/h$

Demand charges: $Demand \times MC = 100MW \times 6.2£/MWh = 620£/h$

Southern region:

$MC_3 = MC_4 \Rightarrow 5 + 0.025 \cdot P_3 = 7 + 0.025 \cdot P_4$

$P_3 + P_4 = 400MW - 50MW$

Hence,

$P_3 = 215MW$

$P_4 = 135MW$

$MC_1 = MC_2 = 10.375£/MWh$

Payments to Generator 3: $P_3 \times MC_3 = 215MW \times 10.375£/MWh = 2230.625£/h$

Payments to Generator 4: $P_4 \times MC_4 = 135MW \times 10.375£/MWh = 1400.625£/h$

Demand charges: $Demand \times MC = 400MW \times 10.375£/MWh = 4150£/h$

Congestion surplus:

$(\pi_{SR} - \pi_{NR}) \times F = (10.375£/MWh - 6.2£/MWh) \times 50MW = 208.75£/h$

Network profit=Network revenue-transmission investment cost.

Network revenue=Congestion surplus=208.75£/h

Transmission Investment Cost

$$= \frac{k \cdot \lambda \cdot F_C}{T_0} = \frac{36 \frac{£}{MW \cdot km \cdot year} \cdot 470km \cdot 50MW}{8760 \frac{h}{year}} = 96.575£/h$$

Hence, the transmission profit is $208.75£/h - 96.575£/h = 112.175£/h$

(ii) We need to find the optimal dispatch for the case C= 150MW (150MW moving from Northern region to Southern region).

Northern region:

$MC_1 = MC_2 \Rightarrow 4 + 0.02 \cdot P_1 = 5 + 0.03 \cdot P_2$

$P_1 + P_2 = 100MW + 150MW$

Hence,

$P_1 = 170MW$

$P_2 = 80MW$

$MC_1 = MC_2 = 7.4£/MWh$

Payments to Generator 1: $P_1 \times MC_1 = 170MW \times 7.4£/MWh = 1258£/h$

Payments to Generator 2: $P_2 \times MC_2 = 80MW \times 7.4£/MWh = 592£/h$

Demand charges: $Demand \times MC = 100MW \times 7.4£/MWh = 740£/h$

Southern region:

$MC_3 = MC_4 \Rightarrow 5 + 0.025 \cdot P_3 = 7 + 0.025 \cdot P_4$

$P_3 + P_4 = 400MW - 150MW$

Hence,

Answers

$P_3 = 165MW$

$P_4 = 85MW$

$MC_1 = MC_2 = 9.125£/MWh$

Payments to Generator 3: $P_3 \times MC_3 = 165MW \times 9.125£/MWh = 1505.625£/h$

Payments to Generator 4: $P_4 \times MC_4 = 85MW \times 9.125£/MWh = 775.625£/h$

Demand charges: $Demand \times MC = 400MW \times 9.125£/MWh = 3650£/h$

Congestion surplus:

$(\pi_{SR} - \pi_{NR}) \times F = (9.125£/MWh - 7.4£/MWh) \times 150MW = 258.75£/h$

Network profit=Network revenue-transmission investment cost.

Network revenue=Congestion surplus=258.75£/h

Transmission Investment Cost

$$= \frac{k \cdot \lambda \cdot F_C}{T_0} = \frac{36 \frac{£}{MW \cdot km \cdot year} \cdot 470km \cdot 150MW}{8760 \frac{h}{year}} = 289.726£/h$$

Hence, the transmission profit is $258.75£/h - 289.726£/h = -30.976£/h$

c.

(i) The transmission company wishes to maximize its profit:

$\max\{\text{Revenue}(F) - Cost(F)\} wrt F.$

$\text{Revenue}(F) = \pi_T(F) \times F$

Marginal Value of transmission: $\pi_T(F) = \pi_{SR}(F) - \pi_{NR}(F)$

$\pi_T(F) = MC_3 - MC_1 = 5 + 0.025 \cdot P_3 - (4 + 0.02 \cdot P_1) = 1 + 0.025 \cdot P_3 - 0.02 \cdot P_1$

We can also use that:

$P_1 + P_2 = D_{NR} + F$

$P_3 + P_4 = D_{SR} - F$

$MC_1 = MC_2$

$MC_3 = MC_4$

Hence,

$P_1 = 80 + 0.6F$

$P_2 = \frac{2}{3} P_1 - \frac{100}{3}$

$P_3 = 240 - \frac{F}{2}$

$P_4 = P_3 - 80$

and by using the above we have the expression: $\pi_T(F) \cdot F = (5.4 - 0.0245F) \cdot F$

$Cost(F) = \frac{k \cdot \lambda \cdot F}{T_0} = \frac{36 \cdot 470 \cdot F}{8760} = 1.932F$

$\text{Profit}(F) = \text{Revenue}(F) - Cost(F) = -0.0245F^2 + 3.468F$

Answers

In order to find the maximum of this equation, we set its derivative = 0.

$$\frac{d\,\mathrm{Pr}\,ofit(F)}{dF} = 0 \Rightarrow 3.468 - 2 \cdot 0.0245F = 0 \Rightarrow F = 70.77MW$$

(ii) The capacity maximizing the benefit of transmission for the entire country satisfies the fact that:
Value of transmission=Marginal investment cost of transmission
In our case:

Marginal investment cost of transmission: $\frac{k \cdot \lambda}{T_0} = \frac{36 \cdot 470}{8760} = 1.932 \frac{£}{MW \cdot h}$

From before we can use that $\pi_T(F) = 5.4 - 0.0245F$

Hence, $5.4 - 0.0245F = 1.932 \Rightarrow F = 141.55MW$

(iii) Generators G1 and G2 and Demand in the south region do not want to be constrained by the transmission link. This means:
MC1=MC2=MC3=MC4

$$MC1 = MC2 \Rightarrow 4 + 0.02P_1 = 5 + 0.03P_2 \Rightarrow P_2 = \frac{2}{3}P_1 - \frac{100}{3}$$

$$MC1 = MC3 \Rightarrow 4 + 0.02P_1 = 5 + 0.025P_3 \Rightarrow P_3 = 0.8P_1 - 40$$

$$MC1 = MC4 \Rightarrow 4 + 0.02P_1 = 7 + 0.025P_4 \Rightarrow P_4 = 0.8P_1 - 120$$

We also know that P1+P2+P3+P4=Demand_north+Demand_south

$$P_1 + \frac{2}{3}P_1 - \frac{100}{3} + 0.8P_1 - 40 + 0.8P_1 - 120 = 500 \Rightarrow P_1 = 212.245MW$$

$$P_2 = \frac{2}{3}P_1 - \frac{100}{3} = 108.163MW$$

$$F = P_1 + P_2 - D_{north} = 220.408MW$$

This means that the optimal capacity from the perspective of a company formed of generators in the North and Demand in the south is 220MW.

Answers